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Given their chaotic nature, how can we know when and where wildfires will occur? That is a question that has puzzled wildfire forecasters for decades and now, thanks to advances in machine learning, we are one step closer to answering it, although we still have some way to go.

The topic of wildfire forecasting is not a new one. However, in recent years media attention around the subject has grown as unprecedented wildfire seasons in Australia (2019/2020) and Canada (2023) have resulted in widespread devastation of local ecosystems and communities.

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Traditional fire forecasting

For nearly half a century, fire danger forecasts have relied on a method that links weather conditions with fire activity to create an index of fire risk, with the Canadian Fire Weather Index (FWI) being the most widely used. However, this approach has its limitations.

A fire needs fuel, and for wildfires that fuel is both living and dead vegetation. The abundance and arrangement of that fuel is known as the 'fuel bed'. If all else is equal, the drier the fuel bed, the higher the fire risk. The FWI primarily estimates the state of fuel moisture based on meteorological conditions affecting a predetermined *typical* Canadian forest fuel bed. However, a *typical* fuel bed does not capture controls such as the moisture levels in living vegetation, the composition of vegetation types, or the actual abundance of available fuel. Consequently, the FWI tends to overestimate fire risk in areas with limited fuel. Moreover, because the FWI was originally developed for Canadian forests, its applicability becomes complex when extrapolated to different ecosystems.

These shortcomings highlight the need for more advanced forecasting techniques to accurately assess wildfire risk across diverse landscapes.

All this is before we even consider the factors that might start a wildfire in the first place (known as ignitions). Ninety per cent of ignitions are caused by the unpredictable behaviour of humans, making them chaotic in nature and hard to predict.

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us to perform global 1 km forecasts.

The foundation of PoF lies in its utilisation of diverse datasets, including information from the ECMWF Integrated Forecasting System (IFS), land cover data, and a newly developed fuel characteristic model.

The training of PoF is made possible thanks to the wealth of historic observations of active fires from satellites. The model mimics what it anticipates satellites will detect in the next few days. Consequently, it could only ever hope to perform as well as the satellite, which emphasises the importance of accurate satellite data for training. The satellite used, MODIS, captures snapshots of active fires once daily, potentially missing fires burning at different times throughout the day, as well as small fires and those obscured by cloud cover.

A promising avenue for enhancing PoF lies in training it with observations from multiple satellite sensors which represent a broader coverage and improved accuracy.

Canadian wildfires in 2023

In 2023, Canada experienced a devastating wildfire season that emphasised the importance of accurate fire forecasting. The wildfires, fuelled by adverse weather conditions and human activities, posed significant challenges to local communities and ecosystems.

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asked the question, can PoF perform under conditions it was not trained for?

Our observations revealed that both versions of the PoF forecast, a standard 9 km version and an experimental 1 km version, delivered accurate predictions of extreme fire activity for the Canadian fires, offering valuable insights up to ten days in advance (see Figures 1 and 2). Whilst PoF accurately forecasts the outbreak of fire activity on 15 May, it does not manage to accurately capture the persistence of activity caused by wildfire spread over the subsequent days.

This is an area where the model will be developed in the future. The timing and location of the onset of extreme wildfire activity in the forecast shows the capability of PoF. These results highlight a step forward in our ability to forecast wildfire occurrence and provide a useful tool for fire management strategies.

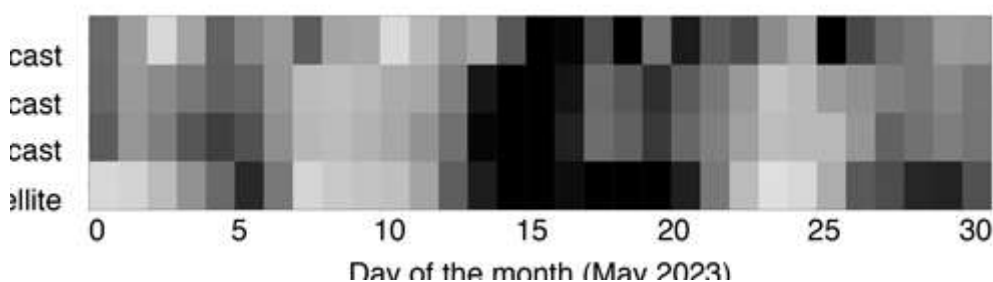


Figure 1: The number of active fires predicted at three different forecast lead times compared with observations from the MODIS satellite in western Canada in May 2023.

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Figure 2: Fires observed by the media satellite on 10 May 2023 (red areas, top left) when fire activity peaked, compared with 1 km forecasts at lead times of 1, 3 and 10 days. The colour bar indicates the probability of fire from the PoF model.

Does the weather care about fires?

Absolutely! Fires have a significant and multifaceted impact on regional weather patterns. One of the many ways fires influence weather is through the release of heat and moisture into the atmosphere. This exchange leads to a phenomenon known as pyroconvection, where rising columns of heated air can trigger the formation of pyrocumulus or pyrocumulonimbus clouds, potentially influencing local weather conditions.

Furthermore, the smoke emitted by wildfires contains a mixture of aerosols, particulate matter, and trace gases. These components interact with incoming solar radiation and reflected upward radiation, resulting in a complex interplay of warming and cooling effects depending on factors such as composition and altitude.

Additionally, fires have long-term indirect effects on the landscape. The removal of vegetation reduces the potential for evapotranspiration and the burn scars left on the landscape influence the permeability of the soil. Both result in changes in surface and soil moisture dynamics. The energy exchange with the atmosphere is also influenced by changes in albedo brought about by fire damage.